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Occupation of a set time of random walks

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Abstract: This bachelor's thesis aims to analyse random walks, emphasizing symmetric random walks. We focus mainly on the occupation of a set times. In this thesis, we find the distribution of maxima and minima, the probability of return to the origin, and the probability of first return to the origin. Then we see the arcsine laws and the probability of spending a given amount of time on the positive and the negative side. We generalize this not only for the positive and negative sides but for any given interval. At the end of the thesis, we create statistical tests based on the theoretical distribution derived in the thesis.

Keywords: symmetric random walk, return to the origin, first return to the origin, arcsine laws, time spent on the positive side

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Introduction

This thesis concerns random walks, mainly symmetric random walks. The thesis is primarily inspired by GRIMMETT and STIRZAKER [2001] and FELLER [1968]. Namely the first three chapters are strongly inspired by GRIMMETT and STIRZAKER [2001]. The rest of the thesis is based on FELLER [1968].

In the first chapter, we define a univariate simple random walk, not necessarily symmetric, and state a few fundamental theorems concerning random walks that will simplify calculations in the rest of the thesis.

In the second chapter, we restrict ourselves only to symmetric random walks and use the fact that this assumption dramatically simplifies all calculations. We define the number of possible paths and the number of possible paths that visit some point. Then we calculate the number of possible paths and the number of possible paths that do not visit the origin.

In the third chapter, we define maxima and minima of a random walk and calculate its distribution. Then we define reaching a new maximum at a particular time and calculate its probability. Lastly, we recognize that the probability of reaching a new minimum is the same as the probability of reaching a new maximum for symmetric random walks.

The fourth chapter concerns returns to the origin. We start by defining return to the origin and first return to the origin. After that, we find out that the probability of no return to the origin up to a specific time is the same as the probability of return to the origin simultaneously. We will then use this theorem in the last chapter for statistical tests. Then we discover a few more interesting facts combining the probability of returning and not returning to the origin. The last part of this chapter deals with first returns to the origin.

The fifth chapter talks about arcsine laws. We state the arcsine law for last visits and after that define the time spent on the positive and negative sides. We prove the arcsine law for sojourn times and define the time spent in an interval and calculate the probability of spending a given amount of time in this interval. The last statement of this chapter concerns the probability of spending a given amount of time on the positive side and returning to the origin.

The last chapter deals with statistical tests. We use the theoretical distribution from the chapter 4. We simulate some alternatives and see if our test can recognize them.

List of abbreviations, assumptions and used notation

In the whole thesis binomial coefficients follow the rule:

$$\binom{n}{a} = 0 \text{ if } a \notin \{0, 1, \dots, n\}$$

$\stackrel{Ta}{=}$ states that the equality comes from the theorem a .

$\stackrel{La}{=}$ states that the equality comes from the lemma a .

$\stackrel{Da}{=}$ states that the equality comes from the definition a .

$\stackrel{LTP}{=}$ states that the equality comes from the Law of total probability.

$\stackrel{IA}{=}$ states that the equality comes from an induction assumption.

$\stackrel{GPR}{=}$ states that the equality comes from the general product rule.

$X \perp\!\!\!\perp Y$ means that X and Y are stochastically independent.

Sets are stated in square brackets i.e. $[S_1 = 1]$.

Set $\mathbb{N}$ consists of all natural numbers (0 is excluded), i.e. $\{1, 2, 3, \dots\}$.

Set $\mathbb{N}_0$ consists of all natural numbers and 0, i.e. $\{0, 1, 2, \dots\}$.

Set $\mathbb{Z}$ consists of all whole numbers, i.e. $\{\dots, -2, -1, 0, 1, 2, \dots\}$.

1. Basic definitions

Remark. First, let us properly introduce what a random walk is. After stating some basic definitions, we will move to the core of this thesis to explore the properties of occupation of a set times.

Definition 1 (Simple random walk in $\mathbb{Z}$)

Let $\{X_n\}_{n=0}^{+\infty}$ be a sequence of independent and identically distributed $\{\pm 1\}$ -valued random variables, that $\forall n \in \mathbb{N}$ and for some $p \in (0, 1)$ satisfy $\mathbf{P}(X_n = 1) = p$ and $\mathbf{P}(X_n = -1) = 1 - p$ which will be denoted by q .

Let $S_0 \in \mathbb{Z}$ and $S_n := S_0 + \sum_{i=1}^n X_i$, then we call the triplet $(\{S_n\}_{n=0}^{+\infty}, p, S_0)$ a simple random walk in $\mathbb{Z}$.

If $S_0 = 0$ we drop it and use the notation of $(\{S_n\}_{n=0}^{+\infty}, p)$. Furthermore, if $p = q = \frac{1}{2}$ we also drop the element p and we call it a symmetric simple random walk in $\mathbb{Z}$.

S_0 is deterministic. If it takes the value of $a \in \mathbb{Z}$, we write the probability in the following way $\mathbf{P}_a(\cdot)$. If $S_0 = 0$ we omit the lower index a and use following notation $\mathbf{P}(\cdot)$.

Remark. We are introducing q as $1 - p$ just for simplicity because writing the term $(1 - p)$ can make calculations very lengthy and confusing. We often refer to n as time, X_i as the i -th step, and S_n as the position of the walk in time n or after n steps. While talking about a simple random walk in $\mathbb{Z}$, we say that the i -th step was rightwards or upwards if $X_i = +1$ and that the i -th step was leftwards or downwards if $X_i = -1$.

Remark. The essential characteristic of random walks is the probability of being in position x in time n . To calculate such a probability, we have to find possible positions.

For example, the random walk can't be in position x in time n if $x > n$ simply because there have not been enough steps to make it up to x . It is also impossible to be in an odd-numbered position after an even number of steps and vice versa. Thus we now define the set of possible positions.

Definition 2 (Set of possible positions)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk. We call the set $A_n = \{z \in \mathbb{Z}; |z| \leq n, \frac{z+n}{2} \in \mathbb{Z}\}$ set of all possible positions of random walk $(\{S_n\}_{n=0}^{+\infty}, p)$ in time n .

Theorem 1 (Probability of position x in time n)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk then for all $n \in \mathbb{N}$

$$\mathbf{P}(S_n = x) = \begin{cases} \binom{n}{\frac{n+x}{2}} p^{\frac{n+x}{2}} q^{\frac{n-x}{2}} & \text{for } x \in A_n, \\ 0, & \text{for } x \notin A_n. \end{cases}$$

Proof. Consider random variables $X_i^+ := \mathbf{1}_{[X_i=1]}$, and $X_i^- := \mathbf{1}_{[X_i=-1]}$, and define new random variables $R_n = \sum_{i=1}^n X_i^+$, $L_n = \sum_{i=1}^n X_i^-$. We can easily deduce that $\forall i \in \mathbb{N}$ the following equality holds: $X_i = X_i^+ - X_i^-$ and thus $S_n = \sum_{i=1}^n X_i =$

$\sum_{i=1}^n (X_i^+ - X_i^-) = \sum_{i=1}^n X_i^+ - \sum_{i=1}^n X_i^- = R_n - L_n$. R_n is the number of rightwards steps and L_n is the number of leftwards steps. It is obvious that the sum of leftwards and rightwards steps has to add up to n , thus we get that $R_n + L_n = n$. Now by adding these two equalities we get $2R_n = S_n + n$ and consequently $R_n = \frac{S_n + n}{2}$.

Clearly, X_i^+ has Bernoulli distribution that has probability of success of p . Hence, R_n as a sum of independent and identically distributed Bernoulli random variables has binomial distribution with parameters n and p . Therefore we get $P(R_n = x) = \binom{n}{x} p^x q^{n-x}$, $x \in \{0, 1, 2, \dots, n\}$. Finally, for $x \in A_n$ we get

$$P(S_n = x) = P\left(R_n = \frac{x+n}{2}\right) = \binom{n}{\frac{x+n}{2}} p^{\frac{n+x}{2}} q^{\frac{n-x}{2}}.$$

□

Remark. We have just proven the most critical theorem of simple random walks. In the rest of this chapter, we will prove three simple lemmata that simplify many calculations in the whole thesis. After that, we will focus more on the thesis goals.

Lemma 2 (Spatial homogeneity)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk and $n \in \mathbb{N}$ then for all $a, j \in \mathbb{Z}$

$$P_a(S_n = j) = P(S_n = j + a).$$

Proof. For any $j, a \in \mathbb{Z}$ holds

$$P(S_n = j) \stackrel{D1}{=} P\left(\sum_{i=1}^n X_i = j\right) = P\left(a + \sum_{i=1}^n X_i = j + a\right) = P_a(S_n = j + a).$$

□

Lemma 3 (Temporal homogeneity)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk and $n, m \in \mathbb{N}$. Then for all $j \in \mathbb{Z}$

$$P(S_n = j) = P(S_{n+m} = j \mid S_m = 0)$$

Proof. For any $j \in \mathbb{Z}$ and $m \in \mathbb{N}$

$$\begin{aligned} P(S_n = j) &\stackrel{D1}{=} P\left(\sum_{i=1}^n X_i = j\right) = P\left(\sum_{i=m+1}^{m+n} X_i = j\right) \\ &= P(S_{n+m} = j \mid S_m = 0), \end{aligned}$$

where the second equality follows from identical distribution of $\{X_n\}_{n=1}^{+\infty}$. □

Lemma 4 (Markov property)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk, $n, m \in \mathbb{N}$, $n \geq m$, $a_i \in \mathbb{Z}$, $i \in \mathbb{N}$ such that $P(S_0 = a_0, S_1 = a_1, \dots, S_m = a_m) > 0$. Then

$$P_{a_0}(S_n = j \mid S_1 = a_1, \dots, S_m = a_m) = P(S_n = j \mid S_m = a_m).$$

Proof. Because $\{X_n\}_{n=1}^{+\infty}$ is a sequence of independent variables, once S_m is known, then the conditional distribution of S_n depends only on steps

$X_{m+1}, X_{m+2}, \dots, X_n$ and therefore cannot depend on any information concerning values $X_1, X_2, \dots, X_{m-1}$ and accordingly $S_1, S_2, \dots, S_{m-1}$. □

2. Number of possible paths

Remark. Having stated basic definitions and lemmata, we may ask questions about the occupation times. Let $a \in \mathbb{N}_0$. How many steps up to time n does our walk spend above a (in interval $[a, +\infty)$)? Similarly, how many steps does the walk spend in the interval $[-a, +\infty)$? We are not able to answer these questions yet. We need more theorems proven to answer these questions. This chapter will discuss the number of possible paths because it is equivalent to the probabilities if we restrict ourselves to symmetric random walks.

Although we make the definitions as general as possible, we need the assumption of symmetry in almost every statement in the rest of the thesis. Only a few theorems can be generalized and admit non-symmetric random walks.

Definition 3 (Number of possible paths)

Let $n \in \mathbb{N}, a, b \in \mathbb{Z}$. We call $N_n(a, b)$ number of possible paths of random walk $(\{S_n\}_{n=0}^{+\infty}, p)$ from point $(0, a)$ to point (n, b) and $N_n^x(a, b)$ number of possible paths from point $(0, a)$ to point (n, b) that visit point (z, x) for some $z \in \{1, 2, \dots, n\}$.

Remark. Next theorem states the number of paths from the point $(0, a)$ to point (n, b) . We again use the assumption on binomial numbers. The proof follows very similarly to the proof of theorem 1.

Theorem 5 (Number of possible paths)

Let $a, b \in \mathbb{Z}, n \in \mathbb{N}$ then

$$N_n(a, b) = \binom{n}{\frac{1}{2}(n+b-a)}.$$

Proof. Let us choose a path from the point $(0, a)$ to point (n, b) and let α be the number of rightwards steps and β be the number of leftwards steps. Then $\alpha + \beta = n$ and $\alpha - \beta = b - a$. By adding these two equalities we get that $2\alpha = (n + b - a)$ and $\alpha = \frac{1}{2}(n + b - a)$. The number of possible paths is the number of ways of picking α rightwards from n steps. Therefore we get $N_n(a, b) = \binom{n}{\alpha} = \binom{n}{\frac{1}{2}(n+b-a)}$. $\square$

Remark. The following theorem is called the Reflection principle because it reflects a part of a given path in the $y = 0$ -axis and uses one-to-one mapping.

Theorem 6 (Reflection principle)

Let $a, b, n \in \mathbb{N}$, then

$$N_n^0(a, b) = N_n(-a, b).$$

Proof. Each path from $(0, -a)$ to (n, b) has to intersect $y = 0$ -axis at least once at some point. Let k be the time of the first intersection with $y = 0$ -axis. By reflecting the segment from $(0, -a)$ to $(k, 0)$ in the x -axis and letting the segment from $(k, 0)$ to (n, b) stay the same as it was, we get a path from point $(0, a)$ to (n, b) which visits 0 in time k . Because reflection is a bijective operation on sets of paths, we get the correspondence between the collections of such paths. $\square$

Remark. During an election where there are only two candidates comes up an interesting question: If candidate A received p votes and candidate B received q votes ($p > q$), what is the probability that A had been strictly ahead of B throughout the count? Answer to this question is provided by the following theorem by setting $b := p - q$ and $n := p + q$.

Theorem 7 (Ballot theorem)

Let $n, b \in \mathbb{N}$. Number of paths from point $(0, 0)$ to point (n, b) that are always positive is equal to $\frac{b}{n} N_n(0, b)$.

Proof. Let us call N the number of paths we are referring to. Because the path ends at point (n, b) and does not visit the origin, the first step has to be rightwards. Thus we now have to subtract the number of paths from $(1, 1)$ to (n, b) that visit the origin from the total number of paths from $(1, 1)$ to (n, b) .

$$N = N_{n-1}(1, b) - N_{n-1}^0(1, b) \stackrel{\text{T6}}{=} N_{n-1}(1, b) - N_{n-1}(-1, b).$$

Hence we get that:

$$\begin{aligned} N &= N_{n-1}(1, b) - N_{n-1}(-1, b) \stackrel{\text{T5}}{=} \binom{n-1}{\frac{n}{2} + \frac{b}{2} - 1} - \binom{n-1}{\frac{n}{2} + \frac{b}{2}} \\ &= \frac{(n-1)!}{\left(\frac{n}{2} + \frac{b}{2} - 1\right)! \left(\frac{n}{2} - \frac{b}{2}\right)!} - \frac{(n-1)!}{\left(\frac{n}{2} + \frac{b}{2}\right)! \left(\frac{n}{2} - \frac{b}{2} - 1\right)!} \\ &= \frac{(n-1)!}{\left(\frac{n}{2} + \frac{b}{2} - 1\right)! \left(\frac{n}{2} - \frac{b}{2}\right) \left(\frac{n}{2} - \frac{b}{2} - 1\right)!} - \frac{(n-1)!}{\left(\frac{n}{2} + \frac{b}{2}\right) \left(\frac{n}{2} + \frac{b}{2} - 1\right)! \left(\frac{n}{2} - \frac{b}{2} - 1\right)!} \\ &= \frac{(n-1)!}{\left(\frac{n}{2} + \frac{b}{2} - 1\right)! \left(\frac{n}{2} - \frac{b}{2} - 1\right)!} \left(\frac{1}{\frac{n}{2} - \frac{b}{2}} - \frac{1}{\frac{n}{2} + \frac{b}{2}} \right) \\ &= \frac{1}{n} \frac{n!}{\left(\frac{n}{2} + \frac{b}{2} - 1\right)! \left(\frac{n}{2} - \frac{b}{2} - 1\right)!} \left(\frac{\left(\frac{n}{2} + \frac{b}{2} - \frac{n}{2} + \frac{b}{2}\right)}{\left(\frac{n}{2} - \frac{b}{2}\right) \left(\frac{n}{2} + \frac{b}{2}\right)} \right) \\ &= \frac{b}{n} \frac{n!}{\left(\frac{n}{2} + \frac{b}{2}\right)! \left(\frac{n}{2} - \frac{b}{2}\right)!} = \frac{b}{n} \binom{n}{\left(\frac{n}{2} + \frac{b}{2}\right)} \stackrel{\text{T5}}{=} \frac{b}{n} N_n(0, b). \end{aligned}$$

□

3. Maxima and minima

Remark. While thinking about our question, we can easily deduce that maximum of the random walk will be very useful. After defining maximum, we will prove a theorem regarding the distribution of the maximum where we use the Reflection principle. Then we define reaching a new maximum and probability of this event. In the last part of this chapter we will state our own lemma that will help with calculations in the thesis. This lemma concerns symmetric random walks and formally prove that the probability of reaching a new maximum is the same as for reaching a new minimum.

Definition 4 (Maximum and minimum)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk. $M_n := \max\{S_i, i \in \{0, 1, \dots, n\}\}$ is called maximum of random walk $(\{S_n\}_{n=0}^{+\infty}, p)$ up to time n and $M_n^- := \min\{S_i, i \in \{0, 1, \dots, n\}\}$ is called minimum of random walk $(\{S_n\}_{n=0}^{+\infty}, p)$ up to time n .

Theorem 8 (Probability of maximum up to time n)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk. Let $r \in \mathbb{N}, b \in \mathbb{Z}$ then

$$\mathbb{P}(M_n \geq r, S_n = b) = \begin{cases} \mathbb{P}(S_n = b) & \text{if } b \geq r, \\ \mathbb{P}(S_n = 2r - b) \left(\frac{q}{p}\right)^{r-b}, & \text{otherwise.} \end{cases}$$

Proof. Obviously, the case when $b \geq r$ is the easier one because we defined M_n as $\max\{S_i, i \in \{0, 1, \dots, n\}\}$ we get that $M_n \geq b \geq r$ therefore $[M_n \geq r] \subset [S_n = b]$ hence we get $\mathbb{P}(M_n \geq r, S_n = b) = \mathbb{P}(S_n = b)$.

Now let $r \geq 0, b < r$. By the theorem 6 $N_n^r(0, b) = N_n(0, 2r - b)$ and thus

$$\begin{aligned} \mathbb{P}(S_n = b, M_n \geq r) &= N_n^r(0, b) p^{\frac{n+b}{2}} q^{\frac{n-b}{2}} = \\ &= N_n(0, 2r - b) p^{\frac{n+(2r-b)}{2}} q^{\frac{n-(2r-b)}{2}} p^{b-r} q^{r-b} \stackrel{\text{T1}}{=} \left(\frac{q}{p}\right)^{r-b} \mathbb{P}(S_n = 2r - b). \end{aligned}$$

□

Definition 5 (Walk reaching a new maximum at particular time)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk and $n, b \in \mathbb{N}$. We say that the walk reached a new maximum b in time n if $M_{n-1} = S_{n-1} = b - 1, S_n = b$. We denote such probability by $\varphi_b(n)$. We say that the walk reached a new minimum $-b$ in time n if $M_{n-1}^- = S_{n-1} = -b + 1, S_n = -b$. We denote such probability by $\varphi_{-b}^-(n)$.

Theorem 9 (Probability of reaching a new maximum b in time n)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk $n, b \in \mathbb{N}$ then

$$\varphi_b(n) = \frac{b}{n} \mathbb{P}(S_n = b).$$

Proof. The main idea of the proof comes from the fact that if we reached a new

maximum in time n , the n -th step had to be upwards. We then use independence.

$$\begin{aligned}
\varphi_b(n) &\stackrel{\text{D5}}{=} \mathbf{P}(M_{n-1} = S_{n-1} = b-1, S_n = b) = \mathbf{P}(M_{n-1} = S_{n-1} = b-1, X_n = +1) \\
&\stackrel{\text{I1}}{=} \mathbf{P}(M_{n-1} = S_{n-1} = b-1) \mathbf{P}(X_n = 1) = p \mathbf{P}(M_{n-1} = S_{n-1} = b-1) \\
&= p(\mathbf{P}(M_{n-1} \geq b-1, S_{n-1} = b-1) - \mathbf{P}(M_{n-1} \geq b, S_{n-1} = b-1)) \\
&\stackrel{\text{T8}}{=} p \left(\mathbf{P}(S_{n-1} = b-1) - \frac{q}{p} \mathbf{P}(S_{n-1} = b+1) \right) \\
&= p \mathbf{P}(S_{n-1} = b-1) - q \mathbf{P}(S_{n-1} = b+1) \\
&\stackrel{\text{T1}}{=} \binom{n-1}{\frac{n}{2} + \frac{b}{2} - 1} p^{\frac{n}{2} + \frac{b}{2}} q^{\frac{n}{2} - \frac{b}{2}} - \binom{n-1}{\frac{n}{2} + \frac{b}{2}} p^{\frac{n}{2} + \frac{b}{2}} q^{\frac{n}{2} - \frac{b}{2}} \\
&= p^{\frac{n}{2} + \frac{b}{2}} q^{\frac{n}{2} - \frac{b}{2}} \left(\frac{(n-1)!}{\left(\frac{n}{2} + \frac{b}{2} - 1\right)! \left(\frac{n}{2} - \frac{b}{2}\right)!} - \frac{(n-1)!}{\left(\frac{n}{2} + \frac{b}{2}\right)! \left(\frac{n}{2} - \frac{b}{2} - 1\right)!} \right) \\
&= p^{\frac{n}{2} + \frac{b}{2}} q^{\frac{n}{2} - \frac{b}{2}} \left(\frac{(n-1)!}{\left(\frac{n}{2} + \frac{b}{2} - 1\right)! \left(\frac{n}{2} - \frac{b}{2} - 1\right)!} \right) \left(\frac{1}{\frac{n}{2} - \frac{b}{2}} - \frac{1}{\frac{n}{2} + \frac{b}{2}} \right) \\
&= p^{\frac{n}{2} + \frac{b}{2}} q^{\frac{n}{2} - \frac{b}{2}} \left(\frac{(n-1)!}{\left(\frac{n}{2} + \frac{b}{2} - 1\right)! \left(\frac{n}{2} - \frac{b}{2} - 1\right)!} \right) \left(\frac{b}{\left(\frac{n}{2} - \frac{b}{2}\right) \left(\frac{n}{2} + \frac{b}{2}\right)} \right) \\
&= p^{\frac{n}{2} + \frac{b}{2}} q^{\frac{n}{2} - \frac{b}{2}} \frac{b}{n} \left(\frac{n!}{\left(\frac{n}{2} + \frac{b}{2}\right)! \left(\frac{n}{2} - \frac{b}{2}\right)!} \right) = \frac{b}{n} \binom{n}{\frac{n}{2} + \frac{b}{2}} p^{\frac{n}{2} + \frac{b}{2}} q^{\frac{n}{2} - \frac{b}{2}} \stackrel{\text{T1}}{=} \frac{b}{n} \mathbf{P}(S_n = b).
\end{aligned}$$

The fourth equality follows from the fact that the event $[M_{n-1} \geq b-1]$ can be split into two disjoint events: $[M_{n-1} \geq b-1] = [M_{n-1} \geq b] \cup [M_{n-1} = b-1]$.

Hence: $\mathbf{P}(M_{n-1} = b-1) = \mathbf{P}(M_{n-1} \geq b-1) - \mathbf{P}(M_{n-1} \geq b)$. The same applies for the probability $\mathbf{P}(M_{n-1} = b-1, S_{n-1} = b-1)$. $\square$

Lemma 10 (Probability of maximum and minimum in symmetric random walks)
Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk $n, b \in \mathbb{N}$ then

$$\varphi_b(n) = \varphi_{-b}^-(n).$$

Proof. The probability of each path of length n is 2^{-n} , therefore, the probability involved depends only on the number of paths. The number of paths of length n such that a new maximum of b is achieved in the last step by reflecting in the $y = 0$ -axis the same as the number of paths of length n that reach a new minimum of $-b$ in the last step. Therefore the probabilities in the statement are the same. $\square$

4. Returns to the origin

Remark. This chapter deals with returns to the origin. We will prove a lemma concerning the difference of two consecutive binomial coefficients. With this lemma, we can prove that the probability of no return up to time $2n$ is the same as that of returning to the origin in time $2n$. After that, we proceed to prove some interesting results where the probability of return to the origin comes up in the formula. Lastly, we calculate the probability of first returns to the origin and state a few lemmata that simplify calculations in the following chapter.

Remark. Following is quite a tricky identity concerning binomial numbers, which may not be evident to the reader at first sight. Therefore we have decided to state it as a lemma.

Lemma 11 (Binomial identity)

Let $m, k \in \mathbb{N}, m > k$ then

$$\binom{m-1}{k} - \binom{m-1}{k-1} = \frac{m-2k}{m} \binom{m}{k}.$$

Proof.

$$\begin{aligned} \binom{m-1}{k} - \binom{m-1}{k-1} &= \frac{(m-1)!}{k!(m-k-1)!} - \frac{(m-1)!}{(k-1)!(m-k)!} \\ &= \frac{(m-1)!}{(k-1)!(m-k-1)!} \left(\frac{1}{k} - \frac{1}{m-k} \right) \\ &= \frac{1}{m} \frac{m!}{(k-1)!(m-k-1)!} \frac{m-2k}{k(m-k)} = \frac{m-2k}{m} \frac{m!}{k!(m-k)!} = \frac{m-2k}{m} \binom{m}{k}. \end{aligned}$$

□

Remark. Following is the definition of return to the origin, which is an essential term for this thesis. Let us come back to our question. While calculating the number of steps the walk spent in the interval $[a, +\infty)$, we find the first passage through a and then set a as a new origin.

Definition 6 (Return to the origin)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk, $k \in \mathbb{N}$. We say a return to the origin occurred in time $2k$ if $S_{2k} = 0$. The probability that in time $2k$ occurred a return to the origin is denoted by u_{2k} . We say that the first return to the origin occurred in time $2k$ if $S_1, S_2, \dots, S_{2k-1} \neq 0$ and $S_{2k} = 0$. The probability that in time $2k$ occurred the first return to the origin is denoted by f_{2k} . We define $f_0 := 0$.

Remark. Thanks to the previous lemma, we can prove the following theorem. After proving mentioned theorem and two useful lemmata, we will finally be able to answer our first question.

Theorem 12 (Probability of no return up to $2n$)

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk, then $\forall n \in \mathbb{N}$:

$$\mathbb{P}(S_1, S_2, \dots, S_{2n} \neq 0) = \mathbb{P}(S_{2n} = 0).$$

Proof.

$$\begin{aligned} \mathbb{P}(S_1, S_2, \dots, S_{2n} \neq 0) &\stackrel{LTP}{=} \sum_{i=-n}^n \mathbb{P}(S_1, S_2, \dots, S_{2n} \neq 0, S_{2n} = 2i) \\ &= 2 \sum_{i=1}^n \mathbb{P}(S_1, S_2, \dots, S_{2n} \neq 0, S_{2n} = 2i) \end{aligned}$$

The probability involved is the probability that the walk does not return to zero and ends in $2i$. Because the random walk is symmetric. We could have restricted ourselves to sum over only positive numbers in the last equality. Also, every path of length $2n$ has the same probability of 2^{-2n} , and thus the probability depends only on the number of paths. Using the theorem 7, we continue with our calculation.

$$\begin{aligned} &= 2 \sum_{i=1}^n \frac{2i}{2n} \binom{2n}{n-i} 2^{-2n} \\ &\stackrel{L11}{=} 2 \cdot 2^{-2n} \sum_{i=1}^n \left(\binom{2n-1}{n-i} - \binom{2n-1}{n-i-1} \right) = 2 \cdot 2^{-2n} \binom{2n-1}{n-1} \\ &= 2^{-2n} \frac{2n}{n} \binom{2n-1}{n-1} = 2^{-2n} \binom{2n}{n} = \mathbb{P}(S_{2n} = 0). \end{aligned}$$

We used the lemma 11 choosing $m = 2n$ and $k = n - i$. The following equality comes from the fact that the sum involved is telescopic and uses the rule about binomial coefficients. $\square$

Remark. As one could think, the probability of no return to the origin can be split into the probability of strictly positive and strictly negative paths, each with probability $\frac{1}{2}$. Next theorem formally proves this.

Lemma 13 (Probability of positive path)

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk, then $\forall n \in \mathbb{N}$:

$$\mathbb{P}(S_1, S_2, \dots, S_{2n} > 0) = \frac{1}{2} \mathbb{P}(S_{2n} = 0) = \mathbb{P}(S_1, S_2, \dots, S_{2n} < 0).$$

Proof.

$$\mathbb{P}(S_1, S_2, \dots, S_{2n} > 0) \stackrel{LTP}{=} \sum_{r=1}^n \mathbb{P}(S_1, S_2, \dots, S_{2n} > 0, S_{2n} = 2r).$$

The r -th term follows equality:

$$\begin{aligned}
& \mathbb{P}(S_1, S_2, \dots, S_{2n} > 0, S_{2n} = 2r) = \mathbb{P}(S_1 = 1, S_2, \dots, S_{2n} > 0, S_{2n} = 2r) \\
& = \mathbb{P}(S_2, S_3, \dots, S_{2n} > 0, S_{2n} = 2r \mid S_1 = 1) \mathbb{P}(S_1 = 1) \\
& = \frac{1}{2} \mathbb{P}(S_2, S_3, \dots, S_{2n} > 0, S_{2n} = 2r \mid S_1 = 1) \\
& \stackrel{*}{=} \frac{1}{2} (\mathbb{P}(S_{2n} = 2r \mid S_1 = 1) - \mathbb{P}(S_2 \cdot S_3 \cdot \dots \cdot S_{2n-1} = 0, S_{2n} = 2r \mid S_1 = 1)) \\
& = \frac{1}{2} (2^{-(2n-1)} N_{2n-1}(1, 2r) - 2^{-(2n-1)} N_{2n-1}^0(1, 2r)) \\
& = \frac{1}{2} 2^{-(2n-1)} (N_{2n-1}(1, 2r) - N_{2n-1}^0(1, 2r)) \\
& \stackrel{\text{T6}}{=} \frac{1}{2} 2^{-(2n-1)} (N_{2n-1}(1, 2r) - N_{2n-1}(-1, 2r)) \\
& = \frac{1}{2} 2^{-(2n-1)} \left(\binom{2n-1}{n+r-1} - \binom{2n-1}{n+r} \right)
\end{aligned}$$

The equality denoted by $*$ comes from the disjoint decomposition:

$$[S_{2n} = 2r] = [S_1 \cdot S_2 \cdot \dots \cdot S_{2n-1} \neq 0, S_{2n} = 2r] \cup [S_1 \cdot S_2 \cdot \dots \cdot S_{2n-1} = 0, S_{2n} = 2r],$$

where $S_1 \cdot S_2 \cdot \dots \cdot S_{2n-1} = 0$ is the shorter way of writing that at least one of $S_1, S_2, \dots, S_{2n-1}$ is equal to 0. When we get back to the original sum, we see that it is indeed a telescopic sum, because the negative parts of r -th terms cancel against the positive parts of $(r+1)$ -st terms and the sum reduces to

$$\begin{aligned}
& \frac{1}{2} 2^{-(2n-1)} \binom{2n-1}{n} = \frac{1}{2} 2^{-2n} \cdot 2 \binom{2n-1}{n} = \frac{1}{2} 2^{-2n} \frac{2n}{n} \binom{2n-1}{n} \\
& = \frac{1}{2} 2^{-2n} \binom{2n}{n} = \frac{1}{2} \mathbb{P}(S_{2n} = 0).
\end{aligned}$$

□

Remark. The following theorem calculates the probability of a non-negative random walk. Interestingly, it is twice the probability of a strictly positive path of symmetric random walk. We use the previous lemma to prove this one easily.

Lemma 14 (Probability of non-negative path)

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk, then $\forall n \in \mathbb{N}$:

$$\mathbb{P}(S_1, S_2, \dots, S_{2n} \geq 0) = \mathbb{P}(S_{2n} = 0) = \mathbb{P}(S_1, S_2, \dots, S_{2n} \leq 0).$$

Proof.

$$\begin{aligned}
& \frac{1}{2} \mathbb{P}(S_{2n} = 0) \stackrel{\text{L13}}{=} \mathbb{P}(S_1, S_2, \dots, S_{2n} > 0) = \mathbb{P}(S_1 = 1, S_2, S_3, \dots, S_{2n} \geq 1) \\
& \stackrel{\text{LTP}}{=} \mathbb{P}(S_2, S_3, \dots, S_{2n} \geq 1 \mid S_1 = 1) \mathbb{P}(S_1 = 1) \\
& = \frac{1}{2} \mathbb{P}(S_2, S_3, \dots, S_{2n} \geq 1 \mid S_1 = 1) \\
& \stackrel{\text{L3}}{=} \frac{1}{2} \mathbb{P}_1(S_1, S_2, \dots, S_{2n-1} \geq 1) \stackrel{\text{L2}}{=} \frac{1}{2} \mathbb{P}(S_1, S_2, \dots, S_{2n-1} \geq 0) \\
& = \frac{1}{2} \mathbb{P}(S_1, S_2, \dots, S_{2n} \geq 0).
\end{aligned}$$

The last equality comes from the fact that

$$[S_1, S_2, \dots, S_{2n-1} \geq 0] = [S_1, S_2, \dots, S_{2n} \geq 0],$$

because if the walk is non-negative in an odd-numbered position, it has to be at least 1 and therefore cannot go below 0 in the next step. $\square$

Remark. The following theorem calculates the probability of position in time combined with not returning to the origin. After that, we will get to the first return to the origin.

Theorem 15 (Probability of position x in time n without returning to the origin)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk and $x \in \mathbb{Z}$ then $\forall n \in \mathbb{N}$:

$$\mathbf{P}(S_1, S_2, \dots, S_n \neq 0, S_n = x) = \frac{|x|}{n} \mathbf{P}(S_n = x).$$

Proof. Let us without loss of generality assume that $x > 0$. The probability involved is equal to the number of paths of length n from the origin to x , that do not return to the origin multiplied by $p^{\frac{n+x}{2}} q^{\frac{n-x}{2}}$ which is the probability of $\frac{n+x}{2}$ rightwards steps and $\frac{n-x}{2}$ leftwards steps. The first term is by Ballot theorem (7) equal to $\frac{x}{n} N_n(0, x)$ and therefore the desired probability is $\frac{x}{n} \mathbf{P}(S_n = x)$. Proof of the case $x < 0$ follows identically. $\square$

Remark. Now let us study the probabilities of first returns. We will prove that for a symmetric random walk, the probability of being strictly positive up to time $2n - 1$ and returning in time $2n$ is the same as the probability of being strictly negative up to time $2n - 1$ and returning in time $2n$. The probability of the first return to the origin can be split into these two probabilities, and from that, we get the result.

Lemma 16 (First return for symmetric random walks)

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk. Then $\forall n \in \mathbb{N}$:

$$\begin{aligned} f_{2n} &= 2 \cdot \mathbf{P}(S_1 > 0, S_2 > 0, \dots, S_{2n-1} > 0, S_{2n} = 0) \\ &= 2 \cdot \mathbf{P}(S_1 < 0, S_2 < 0, \dots, S_{2n-1} < 0, S_{2n} = 0). \end{aligned}$$

Proof.

$$\begin{aligned} f_{2n} &\stackrel{\text{D6}}{=} \mathbf{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_{2n-1} \neq 0, S_{2n} = 0) \\ &= \mathbf{P}(S_1 > 0, S_2 \neq 0, \dots, S_{2n-1} \neq 0, S_{2n} = 0) \\ &\quad + \mathbf{P}(S_1 < 0, S_2 \neq 0, \dots, S_{2n-1} \neq 0, S_{2n} = 0) \\ &= \mathbf{P}(S_1 > 0, S_2 > 0, \dots, S_{2n-1} > 0, S_{2n} = 0) \\ &\quad + \mathbf{P}(S_1 < 0, S_2 < 0, \dots, S_{2n-1} < 0, S_{2n} = 0) \\ &= 2 \cdot \mathbf{P}(S_1 > 0, S_2 > 0, \dots, S_{2n-1} > 0, S_{2n} = 0) \end{aligned}$$

The third equality comes from the fact that if S_1 is positive and $S_2, S_3, \dots, S_{2n-1}$ are non-negative, they all have to be positive as well. The last equality comes from symmetry; therefore, both involved probabilities are the same. $\square$

Remark. Now, we will prove three theorems stating the probability of first returns to the origin as a function of the probability of return to the origin. Then we state a simple lemma that formally proves intuitive thought that the probability of return after $2n$ steps can be split into the product of probabilities of first return after $2r$ steps and then returning in $2n - 2r$ steps.

Theorem 17 (First return as difference of returns)

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk, then $\forall n \in \mathbb{N}$:

$$f_{2n} = u_{2n-2} - u_{2n}.$$

Proof. The event $[S_1, S_2, \dots, S_{2n-1} \neq 0]$ is the union of two disjoint events:

$$[S_1, S_2, \dots, S_{2n-1} \neq 0, S_{2n} = 0] \cup [S_1, S_2, \dots, S_{2n-1} \neq 0, S_{2n} \neq 0].$$

Hence we get

$$\begin{aligned} f_{2n} &= \mathbf{P}(S_1, S_2, \dots, S_{2n-1} \neq 0, S_{2n} = 0) \\ &= \mathbf{P}(S_1, S_2, \dots, S_{2n-1} \neq 0) - \mathbf{P}(S_1, S_2, \dots, S_{2n} \neq 0) \\ &= \mathbf{P}(S_1, S_2, \dots, S_{2n-2} \neq 0) - \mathbf{P}(S_1, S_2, \dots, S_{2n} \neq 0) \stackrel{\text{T12}}{=} u_{2n-2} - u_{2n}. \end{aligned}$$

The second equality comes from the fact that

$$[S_1, S_2, \dots, S_{2n-1} \neq 0] = [S_1, S_2, \dots, S_{2n-2} \neq 0],$$

because an odd-numbered step cannot be equal to 0 and therefore the probability does not depend on this step. $\square$

Lemma 18 (u_{2n-2} in terms of u_{2n})

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk, then $\forall n \in \mathbb{N}$:

$$u_{2n-2} = \frac{2n}{2n-1} u_{2n}.$$

Proof.

$$\begin{aligned} u_{2n-2} &\stackrel{\text{D6}}{=} \mathbf{P}(S_{2n-2} = 0) \stackrel{\text{T1}}{=} 2^{-(2n-2)} \binom{2n-2}{n-1} = 4 \cdot 2^{-2n} \frac{(2n-2)!}{(n-1)!(n-1)!} \\ &= 2^{-2n} \frac{(2n)!}{n!n!} \frac{4n^2}{2n(2n-1)} = 2^{-2n} \binom{2n}{n} \frac{2n}{(2n-1)} \stackrel{\text{T1}}{=} \mathbf{P}(S_{2n} = 0) \frac{2n}{2n-1} \\ &\stackrel{\text{D6}}{=} u_{2n} \frac{2n}{2n-1}. \end{aligned}$$

$\square$

Lemma 19 (f_{2n} in terms of u_{2n})

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk, then $\forall n \in \mathbb{N}$:

$$f_{2n} = \frac{1}{2n-1} u_{2n}.$$

Proof.

$$f_{2n} \stackrel{T17}{=} u_{2n-2} - u_{2n} \stackrel{L18}{=} u_{2n} \frac{2n}{2n-1} - u_{2n} = u_{2n} \left(\frac{2n}{2n-1} - 1 \right) = u_{2n} \frac{1}{2n-1}.$$

□

Lemma 20 (f_{2n} in terms of u_{2n-2})

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk, then $\forall n \in \mathbb{N}$:

$$f_{2n} = \frac{1}{2n} u_{2n-2}.$$

Proof.

$$f_{2n} \stackrel{T17}{=} u_{2n-2} - u_{2n} \stackrel{L18}{=} u_{2n-2} - \frac{2n-1}{2n} u_{2n} = u_{2n-2} \left(1 - \frac{2n-1}{2n} \right) = u_{2n-2} \frac{1}{2n}.$$

□

Lemma 21 (Decomposition of u_{2n})

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk.

$$u_{2n} = \sum_{r=1}^n f_{2r} u_{2n-2r}.$$

Proof.

$$\begin{aligned} \mathbb{P}(S_{2n} = 0) &\stackrel{LTP}{=} \sum_{r=1}^n \mathbb{P}(S_{2n} = 0, S_1, S_2, \dots, S_{2r-1} \neq 0, S_{2r} = 0) \\ &\stackrel{GPR}{=} \sum_{r=1}^n \mathbb{P}(S_{2n} = 0 \mid S_1, S_2, \dots, S_{2r-1} \neq 0, S_{2r} = 0) \cdot \\ &\quad \mathbb{P}(S_1, S_2, \dots, S_{2r-1} \neq 0, S_{2r} = 0) \\ &\stackrel{L4,D6}{=} \sum_{r=1}^n \mathbb{P}(S_{2n} = 0 \mid S_{2r} = 0) f_{2r} \stackrel{L3}{=} \sum_{r=1}^n \mathbb{P}(S_{2n-2r} = 0) f_{2r} \stackrel{D6}{=} \sum_{r=1}^n u_{2n-2r} f_{2r}. \end{aligned}$$

□

5. Arcsine laws

Remark. This chapter consists of several main achievements in the theory of random walks. Arcsine laws for last visits and sojourn times and then equidistribution theorem. It also includes our theorem answering the question, what is the probability of spending a particular time in the interval $[2a, +\infty)$. Arcsine law for last visits follows classical proof stated both in GRIMMETT and STIRZAKER [2001] and FELLER [1968]. We present an alternative proof of the arcsine law for sojourn times and the equidistribution theorem.

Theorem 22 (Arcsine law for last visits)

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk $k, n \in \mathbb{N}, k \leq n$. The probability that up to time $2n$ the last return to origin occurred in time $2k$ is given by

$$u_{2k}u_{2n-2k}.$$

Proof. The probability involved can be rewritten as:

$$\begin{aligned} & \mathbb{P}(S_{2k} = 0, S_{2k+1}, S_{2k+2}, \dots, S_{2n} \neq 0) \\ & \stackrel{GPR}{=} \mathbb{P}(S_{2k+1}, S_{2k+2}, \dots, S_{2n} \neq 0 \mid S_{2k} = 0) \mathbb{P}(S_{2k} = 0) \\ & \stackrel{L3}{=} \mathbb{P}(S_1, S_2, \dots, S_{2n-2k} \neq 0) \mathbb{P}(S_{2k} = 0) \stackrel{T12}{=} \mathbb{P}(S_{2n-2k} = 0) \mathbb{P}(S_{2k} = 0) \\ & \stackrel{D6}{=} u_{2n-2k}u_{2k}. \end{aligned}$$

□

Remark. The following calculations (taken from FELLER [1968]) explain why the theorems are called arcsine. By Stirling formula, we know that $n! \sim \left(\frac{n}{e}\right)^n \sqrt{2\pi n}$ as $n \rightarrow +\infty$, therefore we can continue with calculations by:

$$u_{2k} \stackrel{D6}{=} \mathbb{P}(S_{2k} = 0) \stackrel{T1}{=} \binom{2k}{k} 2^{-2k} = \frac{(2k)!}{(k!)^2} 2^{-2k} \sim \frac{(2k)^{2k} e^{-2k} \sqrt{4\pi k}}{k^{2k} e^{-2k} 2^{2k} 2\pi k} = \frac{1}{\sqrt{\pi k}}.$$

Therefore we get that

$$\alpha_{2n}(2k) \stackrel{T22}{=} u_{2k}u_{2(n-k)} \sim \frac{1}{\pi \sqrt{k(n-k)}}.$$

With some estimations, we can proceed with the following calculation:

$$\mathbb{P}(\alpha_{2n}(2k) \leq 2xn) \sim \sum_{k=0}^{xn} \frac{1}{\pi \sqrt{k(n-k)}} \sim \int_0^{xn} \frac{1}{\pi \sqrt{u(n-u)}} du = \frac{2}{\pi} \arcsin \sqrt{x}$$

Remark. Let us now focus on spending the time on the positive and negative sides. We will introduce a new notation for time spent on the positive and negative sides in the following definition. After that, we will prove the arcsine law for sojourn times which deals with these probabilities.

Definition 7 (Time spent on the positive and negative sides)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk. We say that the walk spent τ time units of n on the positive side if $\sum_{i=1}^n \mathbf{1}_{[S_i > 0 \vee S_{i-1} > 0]} = \tau$. Let T_n denote $\sum_{i=1}^n \mathbf{1}_{[S_i > 0 \vee S_{i-1} > 0]}$ and $\beta_n(\tau)$ denote $\mathbf{P}(T_n = \tau)$. Similarly, we say that the walk spent ζ time units of n on the negative side if $\sum_{i=1}^n \mathbf{1}_{[S_i < 0 \vee S_{i-1} < 0]} = \zeta$.

Remark. The condition $S_i > 0 \vee S_{i-1} > 0$ may be rewritten as the condition $S_i \geq 0 \wedge S_{i-1} \geq 0$.

Remark. The following theorem concerns the probability of spending $2k$ time units of $2n$ on the positive side. We present an alternative proof to the classical one found both in GRIMMETT and STIRZAKER [2001] and FELLER [1968].

Theorem 23 (Arcsine law for sojourn times)

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk and $n \in \mathbb{N}, k \in \{0, 1, \dots, n\}$. Then

$$\beta_{2n}(2k) = u_{2n-2k}u_{2k}.$$

Proof. First, let us start with extreme cases.

$$\begin{aligned} \beta_{2n}(2n) &\stackrel{\text{D7}}{=} \mathbf{P}(T_{2n} = 2n) = \mathbf{P}(S_1, S_2, \dots, S_{2n} \geq 0) \stackrel{\text{L14}}{=} u_{2n} \text{ and} \\ \beta_{2n}(0) &\stackrel{\text{D7}}{=} \mathbf{P}(T_{2n} = 0) = \mathbf{P}(S_1, S_2, \dots, S_{2n} \leq 0) \stackrel{\text{L14}}{=} u_{2n} \end{aligned}$$

Let us fix $n > 1$ and let $1 \leq k \leq n - 1$, then the following equality holds:

$$\begin{aligned}
\beta_{2n}(2k) &\stackrel{D7}{=} \mathbb{P}(T_{2n} = 2k) \stackrel{LTP}{=} \sum_{r=1}^n \mathbb{P}(T_{2n} = 2k, S_1, S_2, \dots, S_{2r-1} \neq 0, S_{2r} = 0) \\
&= \sum_{r=1}^n \mathbb{P}(T_{2n} = 2k, S_1, S_2, \dots, S_{2r-1} < 0, S_{2r} = 0) \\
&\quad + \sum_{r=1}^n \mathbb{P}(T_{2n} = 2k, S_1, S_2, \dots, S_{2r-1} > 0, S_{2r} = 0) \\
&\stackrel{GPR}{=} \sum_{r=1}^n \mathbb{P}(T_{2n} = 2k \mid S_1, S_2, \dots, S_{2r-1} < 0, S_{2r} = 0) \cdot \\
&\quad \mathbb{P}(S_1, S_2, \dots, S_{2r-1} < 0, S_{2r} = 0) \\
&\quad + \sum_{r=1}^n \mathbb{P}(T_{2n} = 2k \mid S_1, S_2, \dots, S_{2r-1} > 0, S_{2r} = 0) \cdot \\
&\quad \mathbb{P}(S_1, S_2, \dots, S_{2r-1} > 0, S_{2r} = 0) \\
&\stackrel{L4}{=} \sum_{r=1}^n \mathbb{P}(T_{2n} = 2k \mid T_{2r} = 0, S_{2r} = 0) \mathbb{P}(S_1, S_2, \dots, S_{2r-1} < 0, S_{2r} = 0) \\
&\quad + \sum_{r=1}^n \mathbb{P}(T_{2n} = 2k \mid T_{2r} = 2r, S_{2r} = 0) \mathbb{P}(S_1, S_2, \dots, S_{2r-1} > 0, S_{2r} = 0) \\
&= \sum_{r=1}^n \mathbb{P} \left(\sum_{i=2r+1}^{2n} \mathbf{1}_{[S_i > 0 \vee S_{i-1} > 0]} = 2k \mid S_{2r} = 0 \right) \frac{1}{2} f_{2r} \\
&\quad + \sum_{r=1}^n \mathbb{P} \left(\sum_{i=2r+1}^{2n} \mathbf{1}_{[S_i > 0 \vee S_{i-1} > 0]} = 2k - 2r \mid S_{2r} = 0 \right) \frac{1}{2} f_{2r} \\
&\stackrel{L3}{=} \sum_{r=1}^n \frac{1}{2} f_{2r} \mathbb{P}(T_{2n-2r} = 2k) + \sum_{r=1}^n \frac{1}{2} f_{2r} \mathbb{P}(T_{2n-2r} = 2k - 2r) \\
&= \sum_{r=1}^n \frac{1}{2} f_{2r} \beta_{2n-2r}(2k) + \sum_{r=1}^n \frac{1}{2} f_{2r} \beta_{2n-2r}(2k - 2r).
\end{aligned}$$

The third equality comes from the disjoint decomposition of

$$[S_1, S_2, \dots, S_{2r-1} \neq 0] = [S_1, S_2, \dots, S_{2r-1} > 0] \cup [S_1, S_2, \dots, S_{2r-1} < 0],$$

And the equality marked $*$ comes from using the condition that up to time $2r$ the steps were on the positive/negative sides, symmetry, and the definition 6.

Now let us proceed by induction. Case for $n = 1$ is trivial because it implies the extreme case. Let the statement be true for all $k \leq n - 1$, then

$$\begin{aligned}
\beta_{2n}(2k) &= \sum_{r=1}^n \frac{1}{2} f_{2r} \beta_{2n-2r}(2k) + \sum_{r=1}^n \frac{1}{2} f_{2r} \beta_{2n-2r}(2k - 2r) \\
&\stackrel{IA}{=} \sum_{r=1}^n \frac{1}{2} f_{2r} u_{2k} u_{2n-2r-2k} + \sum_{r=1}^n \frac{1}{2} f_{2r} u_{2k-2r} u_{2n-2k} \\
&= \frac{1}{2} u_{2k} \sum_{r=1}^n f_{2r} u_{2n-2r-2k} + \frac{1}{2} u_{2n-2k} \sum_{r=1}^n f_{2r} u_{2k-2r} \\
&\stackrel{L21}{=} \frac{1}{2} u_{2n-2k} u_{2k} + \frac{1}{2} u_{2n-2k} u_{2k} = u_{2n-2k} u_{2k}
\end{aligned}$$

□

Remark. Now we will generalize the previous definition and theorem. We will focus on spending time units in the interval $[a, +\infty)$. We prove the theorem that answers our question what the probability of spending a certain amount of time in the interval $[2a, +\infty)$ is. The proof is divided into three parts according to the value of a .

Definition 8 (Time spent above a)

Let $(\{S_n\}_{n=0}^{+\infty}, p)$ be a random walk. Let $n, z \in \mathbb{N}, z \leq n$. We say that the walk spent z time units up to time n in the interval $[a, +\infty)$ (or above a) if $\sum_{i=1}^n \mathbf{1}_{[S_i > a \vee S_{i-1} > a]} = z$. Let $W_n(a)$ denote $\sum_{i=1}^n \mathbf{1}_{[S_i > a \vee S_{i-1} > a]}$.

Remark. The condition $S_i > a \vee S_{i-1} > a$ can again be rewritten as $S_i \geq a \wedge S_{i-1} \geq a$. We can easily see that $\forall n \in \mathbb{N} : W_n(0) = T_n$.

Theorem 24 (Probability of spending $2z$ time units in interval $[2a, +\infty)$)

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk, let $a \in \mathbb{Z}$, then

$$P(W_{2n}(2a) = 2z) = \begin{cases} \sum_{i=1}^n \binom{2(n-i-z)}{n-i-z} \binom{2z}{z} \frac{a}{i} \binom{2i}{i+a} 2^{-2n} & \text{if } a > 0, \\ \binom{2n-2k}{n-k} \binom{2k}{k} 2^{-2n} & \text{if } a = 0, \\ \sum_{i=1}^n \binom{2(n-z)}{n-z} \binom{2(z-i)}{z-i} \frac{a}{i} \binom{2i}{i+a} 2^{-2n} & \text{if } a < 0. \end{cases}$$

Proof. The case $a = 0$ was described in the theorem 23.

Case $a > 0$ follows:

$$\begin{aligned} & P(W_{2n}(2a) = 2z) \\ & \stackrel{GPR}{=} \sum_{i=1}^n P(W_{2n}(2a) = 2z \mid S_1, S_2, \dots, S_{2i-1} < 2a, S_{2i} = 2a) \cdot \\ & P(S_1, S_2, \dots, S_{2i-1} < 2a, S_{2i} = 2a) \\ & \stackrel{D5}{=} \sum_{i=1}^n P(W_{2n}(2a) = 2z \mid S_1, S_2, \dots, S_{2i-1} < 2a, S_{2i} = 2a) \cdot \varphi_{2a}(2i). \end{aligned}$$

Now let us have a look at $P(W_{2n}(2a) = 2z \mid S_1, S_2, \dots, S_{2i-1} < 2a, S_{2i} = 2a)$. Let us now for simplicity define $X^<(2a, 2i)$, which shall denote $[S_1, S_2, \dots, S_{2i-1} < 2a, S_{2i} = 2a]$ and $X^>(2a, 2i)$, which shall denote $[S_1, S_2, \dots, S_{2i-1} > 2a, S_{2i} = 2a]$. Because $a > 0$ following holds

$$\begin{aligned}
& \mathbb{P}(W_{2n}(2a) = 2z \mid X^<(2a, 2i)) \\
& \stackrel{\text{D8}}{=} \mathbb{P}\left(\sum_{k=1}^{2n} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 2z \mid X^<(2a, 2i)\right) \\
& = \mathbb{P}\left(\sum_{k=1}^{2i} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} + \sum_{k=2i+1}^{2n} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 2z \mid X^<(2a, 2i)\right) \\
& = \mathbb{P}\left(0 + \sum_{k=2i+1}^{2n} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 2z \mid X^<(2a, 2i)\right) \\
& \stackrel{\text{L4}}{=} \mathbb{P}\left(\sum_{k=2i+1}^{2n} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 2z \mid S_{2i} = 2a\right) \\
& \stackrel{\text{L2}}{=} \mathbb{P}\left(\sum_{k=2i+1}^{2n} \mathbf{1}_{[S_k > 0 \vee S_{k-1} > 0]} = 2z \mid S_{2i} = 0\right) \\
& \stackrel{\text{L3}}{=} \mathbb{P}\left(\sum_{k=1}^{2n-2i} \mathbf{1}_{[S_k > 0 \vee S_{k-1} > 0]} = 2z\right) \stackrel{\text{D7}}{=} \mathbb{P}(T_{2n-2i} = 2z) \stackrel{\text{D7}}{=} \beta_{2n-2i}(2z).
\end{aligned}$$

The third equality follows the fact that given the condition $[S_1, S_2, \dots, S_{2i-1} < 2a, S_{2i} = 2a]$ then $\sum_{k=1}^{2i} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 0$.

Overall we have that:

$$\begin{aligned}
\mathbb{P}(W_{2n}(2a) = 2z) &= \sum_{i=1}^n \mathbb{P}(T_{2n-2i} = 2z) \cdot \varphi_{2a}(2i) \\
&\stackrel{\text{T9}}{=} \sum_{i=1}^n \mathbb{P}(T_{2n-2i} = 2z) \frac{a}{i} \mathbb{P}(S_{2i} = 2a) \\
&\stackrel{\text{T23}}{=} \sum_{i=1}^n u_{2n-2i-2z} u_{2z} \frac{a}{i} \mathbb{P}(S_{2i} = 2a) \\
&\stackrel{\text{D6}+\text{T1}}{=} \sum_{i=1}^n \binom{2(n-i-z)}{n-i-z} 2^{-2n+2i+2z} \binom{2z}{z} 2^{-2z} \frac{a}{i} \binom{2i}{i+a} 2^{-2i} \\
&= \sum_{i=1}^n \binom{2(n-i-z)}{n-i-z} \binom{2z}{z} \frac{a}{i} \binom{2i}{i+a} 2^{-2n}.
\end{aligned}$$

The case $a < 0$ follows:

$$\begin{aligned}
& \mathbb{P}(W_{2n}(2a) = 2z) \\
& \stackrel{\text{GPR}}{=} \sum_{i=1}^n \mathbb{P}(W_{2n}(2a) = 2z \mid S_1, S_2, \dots, S_{2i-1} > 2a, S_{2i} = 2a) \cdot \\
& \mathbb{P}(S_1, S_2, \dots, S_{2i-1} > 2a, S_{2i} = 2a) \\
& \stackrel{\text{L10}}{=} \sum_{i=1}^n \mathbb{P}(W_{2n}(2a) = 2z \mid S_1, S_2, \dots, S_{2i-1} > 2a, S_{2i} = 2a) \cdot \varphi_{2a}^-(2i).
\end{aligned}$$

$\mathbf{P}(W_{2n}(2a) = 2z \mid S_1, S_2, \dots, S_{2i-1} > 2a, S_{2i} = 2a)$ now follows:

$$\begin{aligned}
& \mathbf{P}(W_{2n}(2a) = 2z \mid S_1, S_2, \dots, S_{2i-1} > 2a, S_{2i} = 2a) \\
& \stackrel{\text{D8}}{=} \mathbf{P}\left(\sum_{k=1}^{2n} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 2z \mid X^>(2a, 2i)\right) \\
& = \mathbf{P}\left(\sum_{k=1}^{2i} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} + \sum_{k=2i+1}^{2n} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 2z \mid X^>(2a, 2i)\right) \\
& = \mathbf{P}\left(2i + \sum_{k=2i+1}^{2n} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 2z \mid X^>(2a, 2i)\right) \\
& \stackrel{\text{L4}}{=} \mathbf{P}\left(\sum_{k=2i+1}^{2n} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 2z - 2i \mid S_{2i} = 2a\right) \\
& \stackrel{\text{L2}}{=} \mathbf{P}\left(\sum_{k=2i+1}^{2n} \mathbf{1}_{[S_k > 0 \vee S_{k-1} > 0]} = 2z - 2i \mid S_{2i} = 0\right) \\
& \stackrel{\text{L3}}{=} \mathbf{P}\left(\sum_{k=1}^{2n-2i} \mathbf{1}_{[S_k > 0 \vee S_{k-1} > 0]} = 2z - 2i\right) \stackrel{\text{D8}}{=} \mathbf{P}(W_{2n-2i} = 2z - 2i) \stackrel{\text{D7}}{=} \beta_{2n-2i}(2z - 2i).
\end{aligned}$$

The third equality follows the fact that given the condition $S_1, S_2, \dots, S_{2i-1} > 2a, S_{2i} = 2a$ then $\sum_{k=1}^{2i} \mathbf{1}_{[S_k > 2a \vee S_{k-1} > 2a]} = 2i$. Therefore overall, we have

$$\begin{aligned}
\mathbf{P}(W_{2n}(2a) = 2z) &= \sum_{i=1}^n \mathbf{P}(W_{2n-2i} = 2z - 2i) \cdot \varphi_{2a}(2i) \\
&\stackrel{\text{T9}}{=} \sum_{i=1}^n \mathbf{P}(W_{2n-2i} = 2z - 2i) \frac{a}{i} \mathbf{P}(S_{2i} = 2a) \\
&\stackrel{\text{T23}}{=} \sum_{i=1}^n u_{2n-2z} u_{2z-2i} \frac{a}{i} \mathbf{P}(S_{2i} = 2a) \\
&\stackrel{\text{D6+T1}}{=} \sum_{i=1}^n \binom{2(n-z)}{n-z} 2^{-2n+2z} \binom{2(z-i)}{z-i} 2^{-2z+2i} \frac{a}{i} \binom{2i}{i+a} 2^{-2i} \\
&= \sum_{i=1}^n \binom{2(n-z)}{n-z} \binom{2(z-i)}{z-i} \frac{a}{i} \binom{2i}{i+a} 2^{-2n}.
\end{aligned}$$

□

Remark. The last theorem in this chapter is the equidistribution theorem. The statement may look similar to the statement of Arcsine law for sojourn times. We only added that the walks return to the origin after $2n$ steps. However, quite surprisingly, we will conclude that the answer to this question is different because it is the same for any k .

Theorem 25 (Equidistribution theorem)

Let $\{S_n\}_{n=0}^{+\infty}$ be a symmetric random walk and $n \in \mathbb{N}$, then $\forall k, l \in \{0, 1, 2, \dots, n\}$:

$$\mathbf{P}(T_{2n} = 2k, S_{2n} = 0) = \frac{u_{2n}}{n+1} = \mathbf{P}(T_{2n} = 2l, S_{2n} = 0).$$

Proof. Let the probability involved in the statement be denoted by $\delta_{2n}(2k)$. We will prove this claim by induction in n .

The case $n = 0$ is trivial, the case $n = 1$ follows:

$$\mathbf{P}(T_2 = 0, S_2 = 0) = \mathbf{P}(S_1 = -1, S_2 = 0) = \frac{1}{4} = \frac{u_2}{2}$$

and consequently

$$\mathbf{P}(T_2 = 2, S_2 = 0) = \mathbf{P}(S_1 = 1, S_2 = 0) = \frac{1}{4} = \frac{u_2}{2}.$$

The last equality comes from the fact that $u_2 \stackrel{\text{D6}}{=} \mathbf{P}(S_2 = 0) \stackrel{\text{T1}}{=} 2^{-2} \binom{2}{1} = \frac{1}{2}$.
Let us now assume that $\forall m \in \{0, 1, \dots, n-1\}$ and $\forall k \in \{0, 1, 2, \dots, m\}$

$$\delta_{2m}(2k) = \frac{u_{2m}}{m+1}$$

$$\begin{aligned} \delta_{2n}(2k) &= \mathbf{P}(T_{2n} = 2k, S_{2n} = 0) \\ &\stackrel{LTP}{=} \sum_{r=1}^n \mathbf{P}(S_1, S_2, \dots, S_{2r-1} \neq 0, S_{2r} = 0, T_{2n} = 2k, S_{2n} = 0) \\ &= \sum_{r=1}^n \mathbf{P}(S_1, S_2, \dots, S_{2r-1} > 0, S_{2r} = 0, T_{2n} = 2k, S_{2n} = 0) \\ &\quad + \sum_{r=1}^n \mathbf{P}(S_1, S_2, \dots, S_{2r-1} < 0, S_{2r} = 0, T_{2n} = 2k, S_{2n} = 0) \end{aligned}$$

Where the last equality comes from the disjoint decomposition:

$$[S_1, S_2, \dots, S_{2r-1} \neq 0] = [S_1, S_2, \dots, S_{2r-1} > 0] \cup [S_1, S_2, \dots, S_{2r-1} < 0].$$

Now we continue in the calculations.

$$\begin{aligned} &\stackrel{GPR}{=} \sum_{r=1}^n \mathbf{P}(T_{2n} = 2k, S_{2n} = 0 \mid S_1, S_2, \dots, S_{2r-1} > 0, S_{2r} = 0) \cdot \\ &\quad \mathbf{P}(S_1, S_2, \dots, S_{2r-1} > 0, S_{2r} = 0) \\ &\quad + \sum_{r=1}^n \mathbf{P}(T_{2n} = 2k, S_{2n} = 0 \mid S_1, S_2, \dots, S_{2r-1} < 0, S_{2r} = 0) \cdot \\ &\quad \mathbf{P}(S_1, S_2, \dots, S_{2r-1} < 0, S_{2r} = 0). \end{aligned}$$

We can now notice that $[S_1, S_2, \dots, S_{2r-1} > 0, S_{2r} = 0] = [T_{2r} = 2r, S_{2r} = 0]$ and $[S_1, S_2, \dots, S_{2r-1} < 0, S_{2r} = 0] = [T_{2r} = 0, S_{2r} = 0]$. Using this fact and lemma

16 we can continue in the calculations

$$\begin{aligned}
&= \frac{1}{2} \sum_{r=1}^n \mathbb{P}(T_{2n} = 2k, S_{2n} = 0 \mid T_{2r} = 2r, S_{2r} = 0) \cdot f_{2r} \\
&+ \frac{1}{2} \sum_{r=1}^n \mathbb{P}(T_{2n} = 2k, S_{2n} = 0 \mid T_{2r} = 0, S_{2r} = 0) \cdot f_{2r} \\
&= \frac{1}{2} \sum_{r=1}^n \mathbb{P}(T_{2n} - T_{2r} = 2k - 2r, S_{2n} = 0 \mid S_{2r} = 0) \cdot f_{2r} \\
&+ \frac{1}{2} \sum_{r=1}^n \mathbb{P}(T_{2n} - T_{2r} = 2k, S_{2n} = 0 \mid S_{2r} = 0) \cdot f_{2r} \\
&\stackrel{L4}{=} \frac{1}{2} \sum_{r=1}^n \mathbb{P}(T_{2n-2r} = 2k - 2r, S_{2n-2r} = 0) \cdot f_{2r} \\
&+ \frac{1}{2} \sum_{r=1}^n \mathbb{P}(T_{2n-2r} = 2k, S_{2n-2r} = 0) \cdot f_{2r}.
\end{aligned}$$

These calculations are correct. However, some terms in the sum are unnecessary because they are equal to zero. The terms in the first sum are positive for $r \in \{1, 2, \dots, k\}$ and in the second sum for $r \in \{1, 2, \dots, n - k\}$. With this adjustment and using notation from the beginning of this proof, we get that:

$$= \frac{1}{2} \sum_{r=1}^k \delta_{2n-2r}(2k - 2r) \cdot f_{2r} + \frac{1}{2} \sum_{r=1}^{n-k} \delta_{2n-2r}(2k) \cdot f_{2r}.$$

By induction assumption, we know that

$$\forall r \in \{1, 2, \dots, n\} \text{ and } \forall k \in \{0, 1, \dots, n - r\} : \delta_{2n-2r}(2k) = \frac{u_{2(n-r)}}{n - r + 1}.$$

Thus we continue our calculations:

$$= \frac{1}{2} \sum_{r=1}^k \frac{u_{2(n-r)}}{n - r + 1} \cdot f_{2r} + \frac{1}{2} \sum_{r=1}^{n-k} \frac{u_{2(n-r)}}{n - r + 1} \cdot f_{2r}.$$

From the lemma 20, we can continue with the calculations as in the following part:

$$= \frac{1}{4} \sum_{r=1}^k \frac{u_{2(n-r)}}{n - r + 1} \frac{u_{2(r-1)}}{r} + \frac{1}{4} \sum_{r=1}^{n-k} \frac{u_{2(n-r)}}{n - r + 1} \frac{u_{2(r-1)}}{r}.$$

Let us now change the counting variable in the second sum to $\rho = n - r + 1$. We get that $r = n + 1 - \rho$, $n - r = \rho - 1$, $r - 1 = n - \rho$, $n - r + 1 = \rho$ and $n - r = \rho - 1$ and the sum goes down from n to $k + 1$, thus we now have:

$$= \frac{1}{4} \sum_{r=1}^k \frac{u_{2(n-r)}}{n - r + 1} \frac{u_{2(r-1)}}{r} + \frac{1}{4} \sum_{\rho=k+1}^n \frac{u_{2(\rho-1)}}{\rho} \frac{u_{2(n-\rho)}}{n - \rho + 1} = \frac{1}{4} \sum_{r=1}^n \frac{u_{2(n-r)}}{n - r + 1} \frac{u_{2(r-1)}}{r}.$$

We can see that the last equality does not depend on k , thus we conclude that

$$\forall k, l \in \{1, 2, \dots, n\} : \delta_{2n}(2k) = \delta_{2n}(2l).$$

The last thing we need to calculate is the probability for all k . We know that there are $n + 1$ possible values for k , thus the result will be in form of $\frac{a}{n+1}$. To calculate a , we use a clever trick:

$$u_{2n} \stackrel{\text{D6}}{=} \mathbf{P}(S_{2n} = 0) \stackrel{LTP}{=} \sum_{r=0}^n \mathbf{P}(T_{2n} = 2r, S_{2n} = 0) = \sum_{r=0}^n \delta_{2n}(2r) = \sum_{r=0}^n \frac{a}{n+1} = a.$$

Thus we may finally finish by saying that

$$\delta_{2n}(2k) = \frac{u_{2n}}{n+1}.$$

□

6. Statistical tests

In this chapter, when we say simply *walk*, we mean a sequence of $\{\pm 1\}$ -valued random variables. Not necessarily independent nor identically distributed. When we want these assumptions to hold, we say *simple random walk*.

From theorem 12 we know that for simple random walks that are also symmetric the following equality holds $\mathbf{P}(S_1, S_2, \dots, S_{2n} \neq 0) = \mathbf{P}(S_{2n} = 0)$. By theorem 1 we also know, again assuming symmetric simple random walk, that $\mathbf{P}(S_{2n} = 0) = 2^{-2n} \binom{2n}{n}$. Let us denote this quantity by $p(n)$. We are going to use this in statistical tests.

Hypothesis: $\{S_n\}_{n=0}^{+\infty}$ is a symmetric simple random walk.

Alternative: $\{S_n\}_{n=0}^{+\infty}$ is not a symmetric simple random walk.

Test statistic:

$$T_m = \sum_{i=1}^m \mathbf{1}_{[S_1^i, S_2^i, \dots, S_{2n}^i \neq 0]}$$

S_j^i denotes the position of i -th independent walk after j steps. This statistic was chosen because it shows how often the walk returns to the origin up to time $2n$ in m independent walks. We know its distribution (under H_0) by theorem 12.

Under H_0 :

$$\mathbf{P}(S_1, S_2, \dots, S_{2n} \neq 0) = p(n)$$

thus

$$\mathbf{1}_{[S_1, S_2, \dots, S_{2n} \neq 0]} \sim \text{Ber}(p(n))$$

therefore we get that

$$T_m \sim \text{Bin}(m, p(n)).$$

Critical region: We reject H_0 at level α if and only if $T_m \geq C_U(\alpha)$ or $T_m \leq C_L(\alpha)$. Where $C_L(\alpha) = \max\{x \in \mathbb{N} : \sum_{j=0}^x \binom{m}{j} p(n)^j (1-p(n))^{m-j} \leq \frac{\alpha}{2}\}$ and $C_U(\alpha) = \min\{x \in \mathbb{N} : \sum_{j=x}^m \binom{m}{j} p(n)^j (1-p(n))^{m-j} \leq \frac{\alpha}{2}\}$.

Power function:

$$\begin{aligned} \beta_n(F) &= \mathbf{P}_F(T_m \in C(\alpha)) = \mathbf{P}_F([T_m \geq C_U(\alpha)] \vee [T_m \leq C_L(\alpha)]) \\ &= \mathbf{P}_F(T_m \geq C_U(\alpha)) + \mathbf{P}_F(T_m \leq C_L(\alpha)) \\ &= 1 - \mathbf{P}_F(T_m < C_U(\alpha)) + \mathbf{P}_F(T_m \leq C_L(\alpha)) \\ &= 1 - \mathbf{P}_F(T_m \leq C_U(\alpha) - 1) + \mathbf{P}_F(T_m \leq C_L(\alpha)) \\ &= 1 - F(C_U(\alpha) - 1) + F(C_L(\alpha)), \end{aligned}$$

where F is the cumulative distribution function of the tested distribution.

P-value:

$$p(m) = 2 \min\{1 - F(t_m - 1), F(t_m)\},$$

where F is the distribution function of $\text{Bin}(m, p(n))$, and t_m is the value of the test statistic T_m .

We will conduct a series of tests with different types of walks and different offsets. We will have four types of walks, varying from simple random walks to more general walks. Walks, where the probability of the next step depends on the parity of the current step to walks, where the probability of the next

steps depends on the previous step and current position. Offset, denoted by the parameter θ , denotes how the probability of each step changes. θ takes values from the set $\{-0.5, -0.49, \dots, 0.49, 0.5\}$.

The values taken for our simulations are:

- The initial probability which we will add to or subtract from our offset to be $\frac{1}{2}$.
- The level of our test is $\alpha = 0.05$.
- The length of our walk is $2n = 10$, therefore we get that $n = 5$.
- The probability used in the binomial distribution therefore is: $p(5) = 0.24609$.
- The number of repetitions in the binomial distribution is $m = 100$.
- Therefore under H_0 we have that the distribution of our test statistic is:

$$T_{100} \sim \text{Bin}(100, 0.24609).$$

- The lower bound of the critical region is $C_L(0.05) = 15$.
- The upper bound of the critical region is $C_U(0.05) = 33$.
- We repeat each test for $r = 1000$ times.
- We calculate the total sum of repetitions that do lie in the critical region:

$$W_{1000} = \sum_{j=1}^{1000} \mathbf{1}_{[15 < T_{100}^j < 33]}.$$

T_{100}^j is the j -th independent repetition of the test statistic T_{100} .

- We will say that the test could not recognize the violation if the assumptions for symmetric random walks are not met but $W_{1000} \leq 50 = r \cdot \alpha$.

The results of our simulations are depicted in the figure 6.1. In the following paragraphs, we will look closely at each violation of the hypothesis.

Simple random walk

For a simple random walk we have that $\mathbf{P}(X_i = 1) = \frac{1}{2} + \theta$, for all $i \in \{1, 2, \dots, n\}$. Figure 6.1 shows us how well our test can recognize the violation of the null hypothesis.

We can see from the figure that for values of θ from -0.05 to 0.05 , we fail to reject H_0 . For any other value, we have more than 50 fails and therefore can rightfully reject the null hypothesis.

Odd/even offsets

For walks where odd and even-numbered steps have different offsets we have that $P(X_i = 1) = \frac{1}{2} + \theta$, for i odd and $P(X_j = 1) = \frac{1}{2} - \theta$, for j even.

Quite surprisingly, the result is very different from the previous one. We see that even for values ranging from -0.24 to 0.23 , we cannot reject H_0 . Therefore, the power of our test against this alternative is very low.

Dependence on the previous step

For the walks when the next step is depending on the previous step we have that $P(X_1 = 1) = \frac{1}{2}$ and then $P(X_i = 1 | X_{i-1} = 1) = \frac{1}{2} - \theta$ and $P(X_i = 1 | X_{i-1} = -1) = \frac{1}{2} + \theta$. Therefore we can see that for $\theta > 0$ the walk should make steps more likely in the opposite direction than in the previous step. For $\theta < 0$, the walk should be more likely to drift into one direction because the probability of the next step having the same value as the previous is bigger than one half.

The figure 6.1 shows that this test has similar results to the test for simple random walks. We fail to reject for values of θ from -0.04 to 0.06 . For these small values, the difference in probability is so small that it is hard to recognize that it is not a simple random walk.

Dependence on current position

The last type of walk is if the probability of the next step being positive depends on the sign of the current position. $P(X_1 = 1) = P(X_i = 1 | S_{i-1} = 0) = \frac{1}{2}$ and then $P(X_i = 1 | S_{i-1} > 0) = \frac{1}{2} - \theta$, $P(X_i = 1 | S_{i-1} < 0) = \frac{1}{2} + \theta$. We see that for $\theta > 0$ the walk should make steps in the direction of $y = 0$ -axis and for $\theta < 0$ the walk should drift into one direction because the probability of the next step to be in the direction of the axis is smaller then one half.

Only for the values of $\theta \in \{-0.01, 0, 0.01, 0.02\}$ we fail to reject the null hypothesis. We have too many failures for any other values, and therefore we can rightfully reject them. The power of this test against this alternative is therefore large.

We can easily see that for $\theta = 0$, all cases result in symmetric simple random walks.

To conclude that, figure 6.1 shows us that the test copes badly with the case when the probability differs for **odd/even**-numbered steps. We cannot reject H_0 even for quite a large value of θ . One possible explanation for this can be that if the walk managed, in one of the first steps, to make the step in the direction that it is less likely to, then the walk can continue going up and down or down and up from that moment on and not return to the origin.

However, for other types of walks, the test performs quite well. Only for a small portion of offsets do we fail to reject H_0 for simple random walks. Similarly, for the walks where the next step depends on the previous one. For the walks where the next step depends on the current position, the test is even better in recognizing the violation.

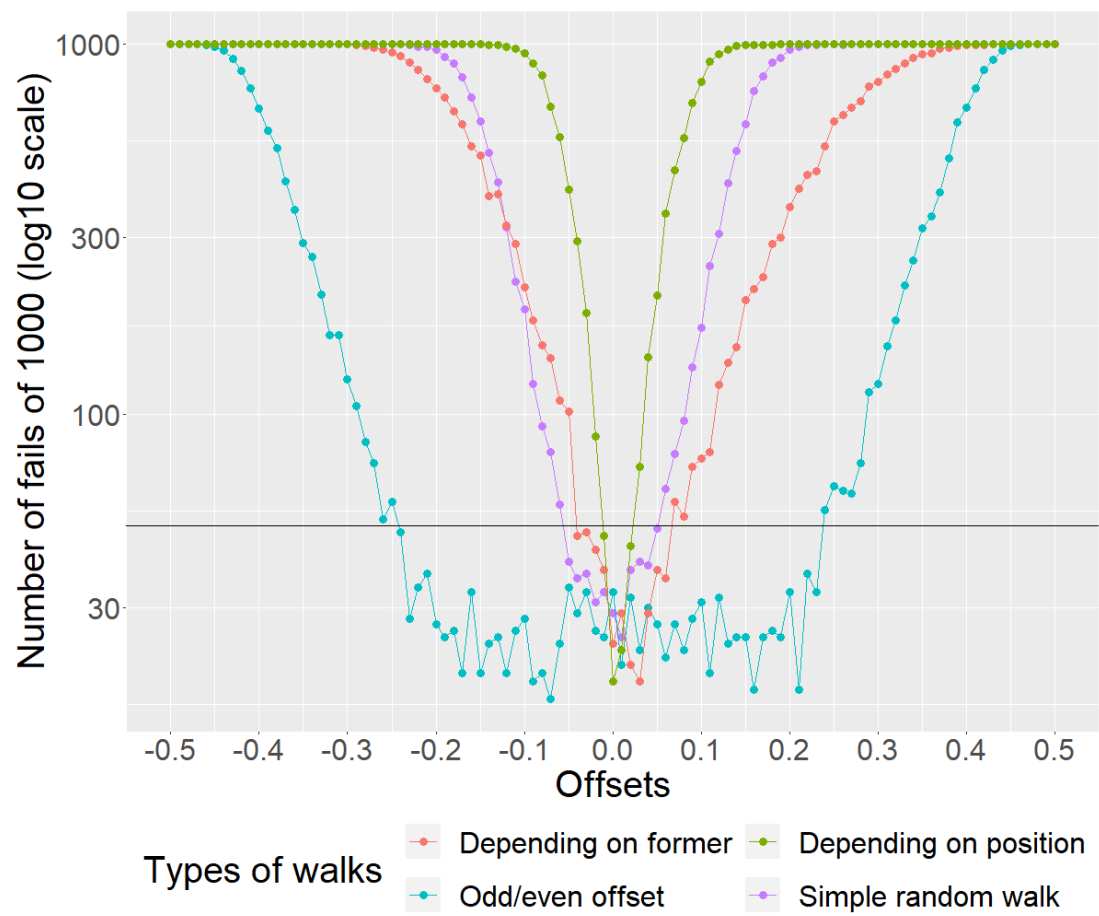


Figure 6.1: Statistical tests

Conclusion

In the thesis, we defined simple random walks and showed some basic properties of simple random walks that can be found in most of the literature concerning random walks. The first chapter was strongly based on GRIMMETT and STIRZAKER [2001]. Mainly because most of the proofs are simple enough that it is hard to come up with different proof. However, all the proofs in the whole thesis are written in our words and mostly in greater detail than in the book.

The second chapter deals with numbers of paths. These calculations are useful for symmetric simple random walks because the probability of rightwards and leftwards steps vanishes and the only term left is the number of paths. This part is again strongly inspired by GRIMMETT and STIRZAKER [2001] and the proofs are again shown in greater details.

In the beginning of the third chapter we defined the minimum and maximum of random walks and proved theorem concerning its distribution. After that we defined what reaching new maximum or minimum means and again calculated its distribution. The main idea of the proof lies in the previous theorem and it is again shown in full detail. Last statement of the chapter is a small lemma on the equality of reaching a new maximum or reaching a new minimum of opposite sign.

The fourth chapter deals with returns to the origin. We proved several theorems, mainly stated in FELLER [1968]. These theorems deal with returns to origin and probability of non-negative and positive paths. One of these theorems was later used for statistical testing in the last chapter. In the latter part of this chapter we proved some useful lemmata that will be used in the fifth chapter mainly dealing with expressing the probability of return to origin in $2n$ steps in terms of return to origin after $2n - 2$ steps and similar equalities.

The fifth chapter is the core part of the thesis. We started with a simple proof of arcsine law for last visits that is similar to the proof stated in FELLER [1968]. We showed why it is named arcsine after that. Then we defined the time spent on the positive and negative side and proved theorem called arcsine law for sojourn times that states the distribution of the time spent on the positive and negative side. We proved this theorem our own way. Following is the previous definition generalized for time spent above a . We then also proved our own way the theorem concerning the distribution of the time spent above a . This proof is a lengthy description of analysis of three cases depending on the positivity or negativity of a . The last theorem of this chapter was theorem stating in what conditions will be the probability of spending $2k$ steps on the positive side equal for all k . The proof of this theorem is again our own.

The last chapter used an equality from chapter 4 for statistical tests. The first part of the chapter dealt with the theory of statistical tests. In the second part we generated many random walks and walks to find out which alternatives can this test easily find out and which are hard to find.

Bibliography

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